

ADDITIONAL MIDTERM 01 REVIEW PROBLEMS – Solutions

STAT 1110: Applied Statistical Concepts and Methods – Fall 2026 with Dr. Ethan P. Marzban

⚠ Caution

These problems aren't necessarily indicative of the exact type of problems that will/may appear on the actual exam.

QUESTION 1

Consider the population of all US citizens and the variable of yearly income.

- (a) Would the distribution of values be left-skewed, right-skewed, or symmetric?

Solution: The distribution would likely be **right-skewed**, because there are only a few very-high earners but more lower earners.

- (b) Which would be larger: the mean or the median?

Solution: The mean would be higher, because the distribution is right-skewed.

- (c) If we removed the 10 individuals with the highest yearly incomes, what would happen to the mean: increase, decrease, or stay the same?

Solution: If we remove large values, the mean would decrease. (You can think of removing large values as reducing the amount of right-skew, toward symmetry.)

QUESTION 2

The probability that a randomly selected card from a standard 52-card deck is a heart is $\frac{1}{4}$. Which statement correctly interprets this probability?

- ☐ If four cards are selected with replacement, at least one will be a heart.

✓ **If one card is selected at random, replaced, and this process is repeated many times, the long-run proportion of selected cards that are hearts will be close to $\frac{1}{4}$.**

QUESTION 3

Imagine we're back in our role as a rock-paper-scissors champion. We've been placed against an opponent who has played "rock" (a "success") in 10 of the 20 rounds we've played. Our research question is: does our opponent seem to favor "rock" over "paper" or "scissors?"

- (a) Describe an observational unit using the context of this study.

Solution: One observational unit is one **trial**; in particular, **one round of rock-paper-scissors**.

- (b) What is the sample size?

Solution: The sample size is **20**, the number of rounds played.

- (c) Define the parameter of interest in words, using the context of the study.

Solution: The parameter of interest is **the long-run proportion of times our opponent plays “rock.”**

- (d) Write down the null and alternative hypotheses using symbols and math, only.

Solution: Defining π_{Rock} as the parameter of interest, the null and alternative hypotheses are:

$$\begin{cases} H_0 : \pi_{\text{Rock}} = 0.333 \\ H_A : \pi_{\text{Rock}} > 0.333 \end{cases}$$

- (e) Suppose we want to set up a simulation of the chance model.

- i. What does one spin of the spinner represent?

Solution: One simulated game of rock-paper-scissors.

- ii. What does the spinner landing on “black” (i.e. a “successful spin”) represent?

Solution: One simulated game of rock-paper-scissors in which our opponent played “rock”

- iii. What should we set as the “probability of ‘success’” in the One Proportion Applet?

Solution: $1/3 \approx 0.333$

- iv. What should we set as the “Number of spins in one repetition” in the One Proportion Applet?

Solution: 20

In addition to the original setup, use the following for parts (f) onwards: 10,000 repetitions of the chance model were conducted using the One Proportion Applet. Shown below are the results of this, with three different proportions highlighted.

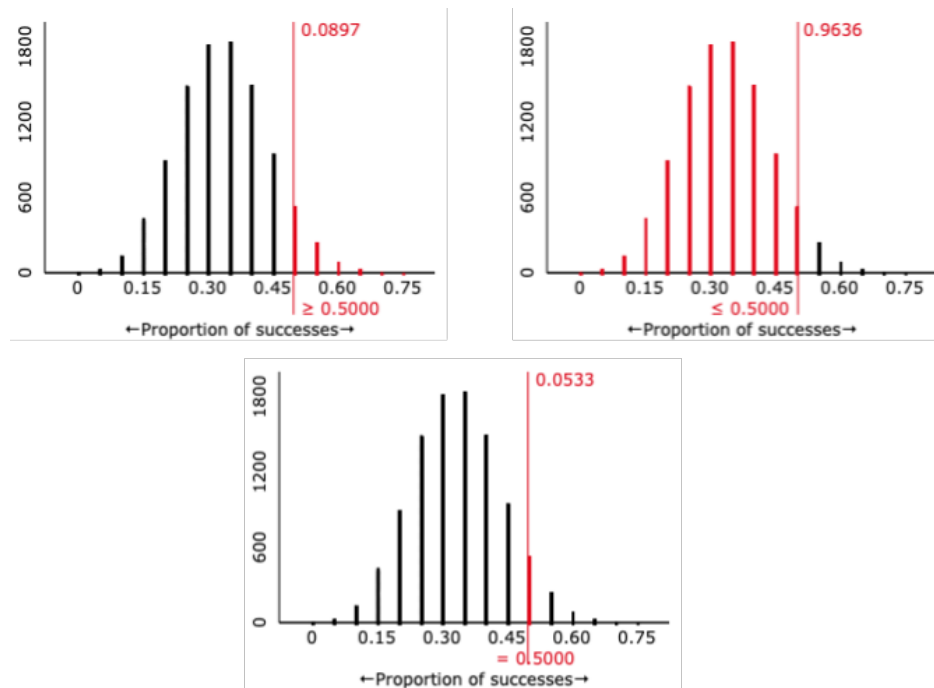


Figure 1: A screenshot of three options from the One Proportion Applet. The highlighted values are: Upper-left: 0.0897; Upper-right: 0.9636; Bottom: 0.0533.

(f) Calculate the approximate p -value.

Solution: Because we have an upper-tailed alternative, we use the upper-left figure to calculate the approximate p -value as **0.0897**

(g) What is the strength of evidence the p -value provides against the null?

- ☐ Not much evidence against the null hypothesis
- ☒ **Moderate evidence against the null hypothesis**
- ☐ Strong evidence against the null hypothesis
- ☐ Very strong evidence against the null hypothesis

Solution: Consult the table from the book (and from Lecture Activity 6)

(h) Summarize the conclusion that you draw from this study based on your analysis. Make sure you explain the reasoning process behind your conclusion by referring to the (1) observed statistic, (2) p -value, (3) justify strength of evidence against the null model (very strong, strong, moderate, weak) based on the p -value, (4) reject/fail to reject the null hypothesis, and (5) overall conclusion in context. Make sure you refer to the **parameter** in your answer.

Solution: The observed statistic is $\hat{p} = 10/20 = 0.50$, and the approximate p -value is 0.0897. This provides moderate evidence against the null hypothesis, that the true long-run proportion of times our opponent plays “rock” is $1/3$. As such, there is moderate evidence to suggest our opponent plays “rock” more often than “paper” or “scissors.”

QUESTION 4

It’s sometimes stated that 80% of all orange cats in the world are male. Tasha, an avid fan of cats, thinks that the true proportion of all orange cats that are male is less than 80%. To test this, Tasha visits her local animal shelter; at her shelter, 36 of the 50 orange cats are male.

- (a) What is the sampling frame?

Solution: The set of all cats at Tasha’s local shelter.

- (b) Define the parameter of interest, in words and in symbols.

Solution: The proportion of all orange cats in the world that are male.

- (c) Write down the null and alternative hypotheses in symbols and math.

Solution: Letting π_{Orange} denote the parameter of interest, then the null and alternative hypotheses are

$$\begin{cases} H_0 : \pi_{\text{Orange}} = 0.8 \\ H_A : \pi_{\text{Orange}} < 0.8 \end{cases}$$

- (d) Explain why we are able to conduct a theory-based approach in this study.

Solution: We are able to conduct a theory-based approach because the validity conditions are met: the number of successes is 36, which is at least 10, and the number of failures is 14, which is also at least 10.

- (e) Calculate the standardized statistic, as indicated by a theory-based approach.

Solution: The hypothesized mean of the null distribution is the null value $\pi_0 = 0.8$, which means the predicted value for SD_{NULL} using a theory-based approach is

$$SD_{\text{NULL}} = \sqrt{\frac{\pi_0(1 - \pi_0)}{n}} = \sqrt{\frac{0.8 \cdot (1 - 0.8)}{50}} \approx 0.0566$$

Therefore, the theory-based standardized statistic is

$$z = \frac{\frac{36}{50} - 0.8}{0.0566} \approx \mathbf{-1.413}$$

- (f) The theory-based approximate p -value is 0.039. Provide an interpretation of this value. (Remember - interpretation is not the same as evaluation!)

Solution: The p -value of 0.039 is the probability of observing a sample proportion (specifically, a sample proportion of 50 orange cats that are male) of male orange cats less than or equal to $(36 / 50) = 72\%$, under the assumption that the true proportion of orange cats that are male is 0.8.